

Student Handbook

A guided route from biological questions to defensible computational evidence

Prepared for Yasin Kaymaz, PhD | Version 1.0 | 14 September 2026

This is a teaching package, not a clinical, legal or institutional accreditation document. Scientific sources are identified by [Sxx] in the Course Source Register.

Start here

Your goal is not to become an expert programmer in one semester. Your goal is to understand what a biological dataset measures, run a small transparent analysis, detect common errors, and explain which conclusions the evidence supports. Reading, checking and explaining code come before writing complex code.

Step	What to do
1	Unzip the Student package and open index.html. Read this handbook and keep the Formula and Audit Sheet nearby.
2	Complete the ungraded diagnostic and Week 00 bridge. Review the sections linked to any missed questions.
3	For each week, read the three outcomes, use the matching slides and open that week's notebook.
4	Use a course pseudonym. Run the notebook from top to bottom, answer the three questions and perform one real check.
5	Restart and run all; save the notebook and JSON summary. Submit only the assignments announced in the grade table.

All notebooks include their own small datasets and require no download of biological data. Colab is the default browser option, but a local Python 3.9+ Jupyter environment or the equivalent paper task is acceptable. No paid AI account, GPU, command line, Git or local package installation is required for core work. Optional plots are skipped when matplotlib is unavailable.

What is essential, and what is not

Essential for the course	Optional next steps
Explain molecules, samples and the experimental unit.	Training a neural network or writing a full bioinformatics pipeline.
Read simple functions, edit a parameter and check a result.	Deriving statistical models or implementing advanced alignment optimizations.
Interpret small tables, basic ratios and model confidence.	Running production-scale RNA-seq, single-cell integration or structure prediction.
Use sources, show uncertainty and protect sensitive data.	Clinical variant classification or authorization to release patient data.

Learning outcomes

ID	By the end you can...	Evidence
LO1	Connect biological questions to DNA, RNA, protein or functional measurements.	D1; midterm; final
LO2	Check sequence files, coordinates, reference versions and sample identities.	Weekly tasks; midterm; final
LO3	Run, explain and test a short provided Python analysis reproducibly.	Weekly tasks; D1/D2; oral
LO4	Justify replication, controls, normalization, uncertainty and multiplicity.	D1; midterm; final
LO5	Evaluate prediction claims using appropriate splits, baselines and evidence limits.	D2; final
LO6	Communicate limitations, sources, AI use and responsible data-sharing decisions.	D1/D2; oral; final

The molecular-biology bridge

Use this section without embarrassment: the class does not assume fluent molecular genetics. The bridge slides provide schematic diagrams, and the diagnostic is not graded.

Cells and molecular levels

An organism contains cells; cells contain many molecules. DNA sequence stores heritable information, RNA can carry or perform functions, and proteins participate in structure, regulation and biochemical activity. DNA, RNA, protein abundance and enzyme activity are different measurement levels. More RNA reads do not automatically mean more active protein. [S01]

Genes, transcripts and proteins

A gene is a genomic region associated with a functional product. Some genes produce noncoding RNAs. A protein-coding gene can yield different transcripts through RNA processing, and those transcripts need not all yield the same protein product. Do not equate the number of genes, transcripts and proteins. [S01,S02]

Exons, introns and translation

A precursor RNA can contain exons and introns. Splicing removes introns and joins retained exon sequences. Not every exon base is part of a protein-coding region. Translation reads an appropriate coding RNA in codons; the reading frame and genetic code matter. An arbitrary genomic interval is not automatically a valid coding sequence. [S01]

Samples, technical repeats and biological replication

A sample is the material measured. An independent biological replicate is an independently sampled or assigned biological experimental unit relevant to the question, such as an independently prepared culture or donor. Repeating a measurement or sequencing an aliquot can assess technical variation without creating a new donor. A paired design can compare conditions within donors, but must retain the pairing. [S10,S12]

Matrices and identifiers

A matrix has rows and columns, but the meaning is determined by the file and software. A bulk count table may have genes in rows and samples in columns; AnnData conventionally places observations in rows and variables in columns. Always label both axes and match sample metadata using unique identifiers. Never attach treatment labels merely by the position of a row. [S10,S13]

Fractions, units and logarithms

A fraction has a numerator and a denominator. GC among known A/C/G/T bases is not the same as GC divided by every character when N is present. Keep bases, reads, read pairs, samples and donors distinct. $\log_2(2)=1$, $\log_2(1)=0$ and $\log_2(0.5)=-1$. A zero denominator needs an explicit decision; quietly adding a number can alter the question.

A worked interpretation

Suppose G1 has 10 raw reads in one sample and 20 in another. The raw ratio is 2, but a biological claim needs library-depth context, correct sample identity, replication and an appropriate analysis. The comparison does not yet show that protein or enzyme activity doubled. A useful answer states both the calculation and its limit. [S10]

Your 14-week route

Week	Focus	Milestone
W00	The biology and numbers bridge	Ungraded bridge + diagnostic
W01	From a biological question to defensible evidence	A1 completion
W02	Files, coordinates and reference versions	A1 completion
W03	Reading and checking short Python programs	A1 completion
W04	Sequence similarity without overclaiming	A1 completion
W05	From sample to reads: designing an experiment	D1 launch
W06	Variation, independence and multiple testing	A1 completion; D1 due
W07	RNA counts, normalization and defensible expression claims	A1 completion

Week	Focus	Milestone
W08	Midterm review and the evidence audit	A1 completion; midterm
W09	Patterns, prediction and data leakage	D2 launch
W10	Single cells and biological replication	A1 completion; D2 due; oral overflow slots
W11	Biological AI and the honest model scorecard	Oral slots + practice
W12	Protein structures: prediction, confidence and function	Oral slots + practice
W13	From a variant to an evidence-based sentence	Oral slots + practice
W14	Responsible bioinformatics and your scientific record	Oral slots + practice

The recurring biological case is a fictional treatment-response experiment. Synthetic examples make assumptions inspectable. The real pasilla excerpt is deliberately only two genes and is used for sample matching, not differential-expression discovery. Real publication and structure-record cards are attributed evidence summaries, not reproduced experiments. [S10,S24,S25]

Assessment and workload

Code	Task	Weight	When
A1	Weekly completion: best 6 of 8	8%	W01,02,03,04,06,07,08,10
A2a	D1: experimental-design project	12%	Launch W05; due end W06
A2b	D2: prediction-evaluation project	12%	Launch W09; due end W10
A3	Midterm, 60 minutes	16%	W08, during lecture
B1	Final, 75 minutes	34%	After W14, scheduled exam period
B2	Individual oral explanation	18%	Scheduled W10-14; project D1 + GENOMES

Weights total 100%. D1 and D2 together are the two core projects; there is no additional compulsory large semester project. Optional extension work and the portfolio compilation do not add a hidden grade component.

Weekly completion: how the 8% works

Each of the eight listed weeks earns a completion credit of 1 for a good-faith submission containing a result or documented failure, substantive attempts at the questions, and an actual check. Incorrect scientific reasoning receives feedback rather than automatic loss of completion credit. An empty template or absent submission earns 0 after the announced correction

opportunity. The best six credits count: $A1 \text{ points} = 8 \times \min(\text{total credits}, 6)/6$. Two missed or incomplete weeks can therefore be dropped. The lecturer sets actual calendar deadlines in the LMS.

Project and exam rules

Projects are individual submissions; discussion in pairs is encouraged, but each student must explain their own choices and acknowledge help. The D1 and D2 briefs specify deliverables and analytic rubrics. Timed exams use the supplied formula sheet and a basic calculator; no internet or AI tools are used during these exams unless a formally approved accommodation changes the arrangement. The individual oral check assesses explanation, not accent, typing speed or memory for software commands.

The 18% oral assessment is an ordinary weighted component in this package. No automatic grade cap, fail rule or oral hurdle is imposed. Institutional rules for attendance, absence, resits, late work, pass thresholds and accommodations must be announced by the course instructor; this package does not invent them.

Planning your time

The source syllabus uses two 45-minute lecture blocks and one 45-minute studio per week. A planning estimate is 31.5 contact hours across 14 weeks, plus a 75-minute final. Allow about 1.5 hours weekly for preparation/review, 6 hours for each project, 1.5 hours for the bridge and about 8 hours of additional exam preparation: roughly 75 hours overall. These are planning assumptions, not measured workloads or a guarantee. Tell the instructor when a core task repeatedly exceeds the intended load. The inherited 3-ECTS administrative description must be checked against the official programme record.

The weekly learning routine

Phase	Your action	Evidence to keep
Predict	Write what you expect before executing.	A short sentence, not a guessed perfect answer.
Run	Execute cells in order; use the provided data.	Result table and relevant runtime information.
Check	Use a small known case, boundary condition or ID match.	What was checked and whether it agreed.
Explain	Name the result, independent unit and limitation.	Three responses in your own words.
Save	Restart/run all, export notebook and JSON.	A reproducible artifact; retain the version submitted.

GENOMES: the audit you can reuse

GENOMES is a course-created mnemonic, not a published validation standard. It is a reminder to inspect common failure points. A reasoned N/A is better than invented metadata.

Letter	Check	What to record
G	Genome / reference / annotation	Accession and version, or justified N/A for non-sequence work.
E	Environment / execution	Python/runtime version, seed, analysis steps and clean rerun.
N	Numbers / dimensions	Rows, columns, samples, features and expected sizes.
O	Offsets / coordinates	0- or 1-based; inclusive or half-open; units and conversions.
M	Multiplicity / experimental unit	What is independent? What family of tests was considered?
E	Entities / identifiers	Unique IDs, sample matching, transcript versions, no silent row-order assumptions.
S	Sources / scope	Provenance, real versus synthetic, citations, permitted use and limitations.

AI use, collaboration and scientific responsibility

AI is optional. During practice and the two projects it may support explanations, debugging and language editing. You remain responsible for checking calculations, identifiers, sources and claims. Do not cite AI output as the original scientific evidence. A transparent no-AI submission is fully acceptable; neither zero AI use nor a check that found no defect is penalized. [S23]

A disclosure can read: "No AI used; I checked the interval length by hand." Or: "Tool and version/date: ...; purpose: explanation of a Python error; content shared: synthetic code only; accepted changes: ...; checks:" You do not need to disclose private unrelated conversations or upload an entire account history.

Do not upload real patient data, identifiable classmate work or confidential research to a public service. Removing names does not by itself establish that genomic data are anonymous. Ask the authorized institutional team before using sensitive data or an unapproved destination. [S20-S22]

A seed, matching output, polished language or a detector score is not proof of misconduct. Concerns should be handled through fair discussion and the institution's process, using actual evidence. Keep a short record of decisions and corrections to make your own reasoning visible.

Help, accessibility and recovery

Situation	Recovery route
No laptop or temporary internet failure	Use the matching paper task in the Lab Workbook; submit a scan/photo or written worksheet.

Situation	Recovery route
Runtime was restarted	Run setup first and then all cells in order. Do not rerun random cells until a desired result appears.
NameError	Check whether the defining cell ran and whether a name was mistyped.
Division by zero / empty input	Inspect the denominator; undefined is sometimes the correct outcome.
AI result seems confident but strange	Check a primary source and a known numerical example. Confidence of wording is not evidence.
Need language or access accommodation	Discuss the approved route with the instructor. Use the plain-text lecture companion, enlarged slides and glossary. Optional complexity is not required.

Readings: a small, purposeful selection

Required reading is the relevant course explanation and selected sections of official documentation, not every full paper in the source register. Spend about 10-20 minutes on a focused section before class. The primary papers are available for evidence checks and optional deeper study.

Weeks	Focused reading
00-01	S01: DNA/RNA/protein foundations; S02: release-specific counting example.
02-04	S03/S04: coordinate conventions; S05: basic Python; S06/S26: alignment and E-values.
05-08	S07: p-value interpretation; S10: count input and sample metadata; S08: BH principle.
09-10	S11: data leakage; S12: biological replication; S13: matrix axes.
11-12	S14/S15: compare stated evaluation scopes; S16: pLDDT and PAE.
13-14	S18: ClinVar review status; S19: categories; S20-S23: responsible data/AI use.